崑山科技大學電子工程系大學部四年制課程科目表

Kun Shan University 4-Year College Curricula (Day Division)

課程名稱(講授時數－實習時數－學分數)

Course Name (Lecture Hours – Practice Hours – Credits)

114.04.08 系課程委員會修訂通過
114.04.22 院課程委員會議修正通過
114.07.22 校課程委員會通過
112 學年度起入學適用
Apply to the freshmen enrolled in 112

第一學年 1st Academic Year		第二學年 2nd Academic Year		第三學年 3rd Academic Year		第四學年 4th Academic Year		
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	
共同必修科目(合計 30 學分) General Education Required Courses (30Credits)								
體育分組(一) Group Physical Education(I) (2-0-2) 中文閱讀與表達(一) Chinese Reading and Expression(I) (2-0-2) 英文聽力與閱讀(一) English Listening and Reading(I) (2-0-2)	體育分組(二) Group Physical Education(II) (2-0-2) 勞作教育(一) The Course of Labor Education(I) (0-1-0.5) 中文閱讀與表達(二) Chinese Reading and Expression(II) (2-0-2) 英文聽力與閱讀(二) English Listening and Reading(II) (2-0-2) 校訂 分類通識 School-designated Liberal Education (2-0-2)	勞作教育(二) (0-1-0.5) 校訂 分類通識 Liberal Education (2-0-2) 院訂 核心通識 科技英文 Scientific and Technological English (2-0-2) 院訂 分類通識 College-designated Liberal Education (2-0-2)	服務學習與公民教育 Public Service and Civic Education (0-1-0) 校訂 分類通識 School-designated Liberal Education (2-0-2)	生涯發展與職業倫理 Career Development and Work Ethics (2-0-2) 院訂 分類通識 College-designated Liberal Education (2-0-2)	校訂核心通識--外語選讀*** Foreign Languages (2-0-2)			英文能力檢定* English Proficiency Examination (1-0-1)
6-0-6	8-1-8.5	6-1-6.5	2-1-2	4-0-4	2-0-2			1-0-1
共同選修科目(合計 1 學分) General Education Selective (1 credit)								
			外語--第二外語 *** (2-0-2)		外語--職場英文 *** (工商 2-0-2)			跨領域學習護照 * Interdisciplinary Learning Passpor (1-0-1)
								(1-0-1)

註：

*「英文能力檢定」依「崑山科技大學學生英文基本能力檢定實施辦法」。"English Proficiency Examination" is enacted according to "Kun Shan University Guidelines for English Proficiency Requirement."

**跨領域學習護照，依「崑山科技大學通識學習護照實施辦法」辦理。"Interdisciplinary Learning Passport" is enacted according to "Kung Shan University Guidelines for General Education Learning Passport. "

***「外語選讀」為選項式必修科目，課程包括「第二外語」及「職場英文」，擇一門修習即可。"Foreign Languages" compromises "Foreign Languages (other than English)" and "Workplace English", among which students choose 1 course. While the courses are offered in the third academic year, "Foreign Languages (other than English)" are given preference to the sophomores.

院訂必修科目(合計 10 學分) Compulsory Courses of College (Total：10 Credits)		
微積分(一)	基本電子學實	工程數學

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(3-0-3) 工程技術概論 (2-0-2)	習 (0-2-2)	(3-0-3)					
5-0-5	0-2-2	3-0-3					
專業必修科目(合計 70 學分) Required subjects for majors (Total：70 Credits)							
數位科技微學程 C 語言程式設計 C programming Design (2-0-2)		數位科技微學程 微算機原理與介面實習 Principle of Microcomputer and Interface Practice (2-2-4)			數位科技微學程 智慧化科技應用實務 The Practice of Intelligent Technology Application (3-0-3)	暑期校外實習 After School Practice (0-*-4)	
普通物理與實驗(一) University Physics Experiment(I) (2-1-3) C 語言程式設計實習 C programming Design Practice (0-1-1) 電路板佈局設計與實習 Print Circuid Board Layout Design and Practice (2-2-4)	微積分(二) Calculus(II) (3-0-3) 普通物理與實驗(二) University Physics Experiment (II) (2-1-3) 數位邏輯設計與實習 Digital Logic Design and Practice (2-1-3) 專業證照丙級輔導 Practical of Professional C Class Certificate (0-2-2)	電路學(一) circuit theory(I) (3-0-3) 電子學與實習(一) Electronics and Practice(I) (2-2-4)	電路學(二) Circuit theory(I) (3-0-3) 電子學與實習(二) Electronics and Practice(II) (2-2-4) 專業證照乙級輔導 Grade B Professional License Counseling (0-2-2)	實務專題(一) Practice Project(I) (0-1-1) 基礎量測及實習 The Fundamental of Measurement and Experiments (2-2-4) 車用晶片模組應用系統概論 Introduction to automotive chip module application system (3-0-3) 車用晶片模組製造生產實務 (2-2-4)	實務專題(二) Practice Project(II) (0-1-1) 學術倫理* Academic ethics (1-0-1) 專業證照實習 Practical of Professional Certificate (0-1-1) 車用晶片模組規劃與設計 (3-0-3) 車用晶片模組測試檢驗實務 (2-2-4)		
(6-4-10)	(7-4-11)	(7-4-11)	(5-4-9)	(7-5-12)	(9-4-13)	(0-*-4)	

註*：「學術倫理」之實施與學分取得依課務組網頁之「學術倫理教育實施要點」辦理。

The implementation of "Academic Ethics" and the acquisition of credits are handled in accordance with the "Implementation Points of Academic Ethics Education" on the website of the Academic Affairs Group.

專業選修科目 Major elective subjects are taught according to the actual conditions of each semester							
				＊技優學程 感測器原理與應用實務 AI Sensor and it's application (1-2-3)	＊技優學程 機電整合實務 Mechatronics Practice (1-2-3) 智慧製造 Intelligent Manufacturing (3-0-3)	＊技優學程 產線智慧化實務(一) Practice on Intelligent Production Line (I) (1-2-3) 智慧生產管理 Intelligent Production Management (3-0-3) 學期校外實習 Term School	＊技優學程 產線智慧化實務(二) Practice on Intelligent Production Line (II) (1-2-3)

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	互動多媒體程式設計 Interactive Multimedia Programming (3-0-3)	電化學* Electrochemistry (3-0-3)	超大型積體電路設計與實習 Very Large Scale Integrated Circuit Design and Practice (2-1-3) 半導體元件與物理 Semiconductor Device and Physics (3-0-3) 數位系統設計與實習 Digital System Design and Practice (2-1-3) 儀表控制軟體與模擬 Instrument Controlling Software and Simulation (2-1-3) 微算機系統設計實務 Practic on Micrcomputer System Design (2-1-3) 儲能系統與產業概論 Introduction to Energy Storage Systems and Industry (3-0-3)	可程式控制器基礎及實習 Programmable Logic Controller Foundation And Practice (2-2-4) 類比電路設計與應用 Analog Circuit Design and Application (3-0-3) 全客戶積體電路佈局設計與實習 Full Custom Integrated Circuit Design and Practice (2-1-3) 積體電路製程 Integrated Circuit Engineering (3-0-3) 通訊原理 Principle of Communications (3-0-3) 機器人智慧化實務與應用 Practice and Application of Robot Intelligence (3-0-3) 儲能系統應用與測試 Design and Application on Energy Storage System (3-0-3) 智慧感測與雲端控制應用 Intelligent Sensing and Cloud Control Applications (3-0-3) 充電樁製造生產實務 Charging pile manufacturing and production practices (2-2-4)	可程式控制器應用及實習 Application and Practice of PLC (2-2-4) 電力電子學與實習 Power Electronics and Practice (2-1-3) 智慧機器人控制系統 Intelligent Robot Control System (3-0-3) 數位積體電路設計與佈局 Digital VLSI Design and Layout (2-1-3) 自動控制 Automatic Control (3-0-3) 嵌入式系統設計與實習 Design and Practice of Embedded System (2-1-3) 智慧車用電子實務 Practice of intelligent Automotive Electronics (1-2-3) 車聯網原理與實作 Principles of Vehicle Intranet and the Laboratory (1-2-3)	Practice (0-*-9) 學年校外實習(一) Annual Internship (I) (0-*-9) 汽機車電子技術 Vehicle Electronic Technology (3-0-3) 人機介面實務 Practice on Human-Machine Interface (3-0-3)	載具節能實務 Carrier Energy Saving Practice (3-0-3) 學期校外實習 Term School Practice (0-*-9) 學年校外實習(二) Annual Internship (II) (0-*-9) 光電半導體製程 Optoelectronics Semiconductor Processing Technolog (3-0-3) 燃料電池 Fuel Cells (3-0-3)
(2-1-3)	(6-0-6)	(6-0-6)	(14-4-18)	(26-5-31)	(23-11-34)	(10-2-30)	(7-2-30)

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最低畢業總學分為 130 學分 The minimum total credits for graduation is 130 credits

智慧運算與應用數位科技微學程(合計 11 學分)

C 語言程式設計 C programming Design Practice (2-0-2)	通識(E) General Education(E) (2-0-2) 微算機原理與介面實習 Principle of Microcomputer and Interface Practice (2-2-4)		智慧化科技應用實務 The Practice of Intelligent Technology Application (3-0-3)				
(2-0-2)	(4-0-6)		(3-0-3)				

學生須修畢上述本系智慧運算與應用數位科技微學程之課程始得畢業。

To be eligible for graduation, students must complete the department's Micro Program in Intelligent Computing and Applied Digital Technology.

綠色科技學程

跨域課程 *綠能技術 (3 學分)	跨域課程 *綠色能源 (3 學分)	跨域課程 *綠色產業 (3 學分)	跨域課程 *綠色環境(3 學分)	跨域課程 *綠色材料 (3 學分)	跨域課程 *智慧綠能 (3 學分)

註 Notes：

1. 最低畢業學分為 130 學分，必修 110 學分、選修 20 學分。
   A minimum of 130 credits is required for graduation, including 110 required credits and 20 elective credits.
2. 有*之科目為工程學院跨系學程－綠色科技學程-學程科目：綠能技術、綠色能源、綠色產業、綠色環境、綠色材料、智慧綠能。其餘為選修課程(外系至少選修 2 門課且須為不同的學程科目課程)，合計修課至少 15 學分以上，成績及格者始認定修習完成本學程。
   Courses marked with an asterisk (*) are part of the College of Engineering's Interdisciplinary Program in Green Technology, which covers the following categories: Green Energy Technology, Green Energy, Green Industry, Green Environment, Green Materials, and Smart Green Energy. Other elective courses require students to take at least two courses from different categories offered by other departments, totaling no fewer than 15 credits. Only students who pass all required courses will be recognized as having completed the program.
3. 課程分(一)(二)(三)(四)者，修課得不受前後段之限制。
   For courses labeled (I), (II), (III), or (IV), students may take them in any order without prerequisite restrictions.
4. 暑期校外實習之實施為**至少** 8 週 320 小時，另外訂定執行辦法。學期校外實習之實施為開設 9 學分以上，至少為期 4.5 個月之校外實習課程，修讀實習課程期間，除依各校訂定定期返校之座談會或研習活動等外，學生應全職於實習機構實習。若於四年級選修學期校外實習課程，則可抵免暑期校外實習及專業選修 7 學分。
   The summer off-campus internship must last at least 8 weeks (320 hours) and will follow separately defined implementation guidelines. The semester-long internship carries 9 or more credits and must last at least 4.5 months. During the internship, except for mandatory return-to-campus seminars or training sessions organized by the university, students are required to work full-time at the host organization. If a senior student enrolls in the semester-long internship course, it may be counted in place of the summer internship and 7 credits of professional electives.
5. 學生須修畢本系智慧運算與應用數位科技微學程(通識教育 E、C 語言程式設計、微算機原理與介面實習、智慧化科技應用實務)課程始得畢業。
   Students must complete the department's Micro Program in Intelligent Computing and Applied Digital Technology (including General Education E, C Language Programming, Microcomputer Principles and Interface Practice, and Practical Applications of Intelligent Technology) to be eligible for graduation.
6. 學生須修畢本系電動車車用晶片模組應用技優學程（車用晶片模組應用系統概論、車用晶片模組規劃與設計、車用晶片模組製造生產實務、車用晶片模組測試檢驗實務)、充電樁開發製造技優學程(充電樁系統概論、充電樁規劃與設計、充電樁製造生產實務、充電樁測試檢驗實務)或本校跨院系學程，始得畢業。
   Students must complete one of the following programs to graduate:
   Advanced Technical Program in Automotive Chip Module Applications (Introduction to Automotive Chip Module Systems, Planning and Design, Manufacturing Practices, and Testing & Inspection), Advanced Technical Program in Charging Pile Development and Manufacturing (Introduction to Charging Pile Systems, Planning and Design,

Manufacturing Practices, and Testing & Inspection), or an approved cross-college interdisciplinary program offered by the university.